

Abhinaya B
Senior Data Engineer
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PROFESSIONAL SUMMARY:

- 9+ years designing data-intensive applications using Apache Spark, Hadoop, and cloud platforms (AWS, Azure) to build real-time analytics, machine learning pipelines, and scalable data warehouses. Expertise spans financial compliance, retail analytics, and healthcare data ecosystems.
- Collaborated with business teams, data stewards, and IT teams to align data governance frameworks with organizational objectives. Streamlined metadata management using Collibra and Alation to enhance data discoverability and user self-service capabilities.
- Developed end-to-end data pipelines in Azure Data Factory for ingesting structured/unstructured data from APIs, IoT devices, and SQL Server. Utilized Azure Databricks for collaborative analytics workspaces and Synapse Analytics for enterprise-grade data warehousing.
- Engineered serverless ETL workflows using AWS Glue and Lambda to process streaming data (Kafka, Kinesis) and batch data (S3, Redshift). Integrated Step Functions for workflow orchestration and CloudWatch for proactive pipeline monitoring.
- Leveraged Spark Core and Spark SQL for distributed data processing, including complex transformations, window functions, and ML lib for feature engineering. Optimized execution plans using Catalyst Optimizer and Tungsten Engine.
- Designed Athena Workgroups with granular access controls and query tagging to align usage with departmental budgets. Implemented S3 lifecycle policies for tiered storage and archival strategies.
- Architected relational and dimensional models in Erwin Data Modeler for financial and retail datasets. Integrated metadata repositories (IBM Meta Data Workbench) with Snowflake for lineage tracking and impact analysis.
- Built SSRS reports and Power BI dashboards to visualize sales performance, supply chain metrics, and customer segmentation. Utilized MDX queries for OLAP cubes and DAX for advanced calculations.
- Automated infrastructure provisioning (Terraform) and configuration management (Ansible) for Hadoop/Spark clusters. Implemented Git workflows for version control and Jenkins for continuous deployment.
- Configured MongoDB sharding for horizontal scalability and DynamoDB global tables for multi-region redundancy. Developed Python scripts to automate Redis cache updates and DynamoDB backups.
- Deployed Hadoop clusters (HDFS, YARN) with Hive for SQL-like querying and Pig for ETL scripting. Optimized Hive tables using ORC formatting and dynamic partitioning for query performance.
- Led Scrum ceremonies (sprint planning, retrospectives) for data engineering teams. Advocated for Test-Driven Development (TDD) to ensure code quality and reduce regression risks.
- Designed RESTful APIs (Python, Flask) to expose CRM and ERP data to internal applications. Implemented OAuth2 security and Swagger documentation for developer accessibility.
- Built Spark Streaming pipelines with Kafka for real-time fraud detection and customer behavior analysis. Utilized stateful processing and watermarking for event-time aggregations.
- Migrated on-premises SQL Server and Oracle databases to Azure Synapse and Delta Lake, modernizing schemas for cloud compatibility and ACID transaction support.
- Configured AWS IAM roles and policies for least-privilege access. Implemented Azure AD authentication for Synapse and Databricks, ensuring GDPR compliance via encryption and audit trails.
- Architected fault-tolerant data pipelines using Spark RDDs and Hive LLAP for concurrent querying. Designed partitioning strategies for parallel processing of large-scale datasets.
- Connected AWS Glue crawlers with Azure Data Lake Storage for cross-cloud metadata synchronization. Enabled hybrid querying via Databricks Unity Catalog and Redshift Spectrum.
- Experience of Partitions, bucketing concepts in Hive and designed both Managed and External tables in Hive to optimize performance. Experience with different file formats like Avro, parquet, ORC, Json and XML.

TECHNICAL SKILLS:

Operating Systems: Windows 98/2000/XP/7,8, Mac OS and Linux CentOS
Programming Languages: Python, R
Web Technologies: XML, JSON
IDE: PyCharm, Jupyter, PyStudio, Sublime Text, Visual Studio Code
Azure Services: Azure Data Lake, Azure Databricks, Azure Synapse Analytics, Azure Data Factory, Azure Functions, and Azure Blob Storage, Azure Data Lake Storage
AWS Services: Amazon EC2, Amazon S3, Amazon DynamoDB, Amazon Glue, Amazon Simple DB, Amazon MQ, Amazon ECS, Amazon Lambdas, Amazon RDS, Amazon CloudWatch, Amazon SQS, AWS Identity and access management
Databases/Servers: MySQL, Redis, SFTP, FTP, PostgreSQL, Aurora DB, DynamoDB, MongoDB
Big Data Technologies: Hadoop, MapReduce, HDFS, Sqoop, PIG, Hive, HBase, Kafka, Yarn, Apache Spark
ETL: Redshift, Postgres-SQL, Snowflake
Web Services/ Protocols: TCP/IP, UDP, FTP, HTTP/HTTPS, SOAP, Rest, Restful

PROFESSIONAL EXPERIENCE:

Innovate Memphis- Memphis, TN
Senior Data Engineer

Sept 2023 - Present

Responsibilities:

- Applied machine learning techniques (Decision Trees, Linear/Logistic Regression) and statistical modeling to enhance predictive analytics and support data-driven decision-making.
- Designed and deployed end-to-end BI solutions on Azure using Data Factory, Data Lake Analytics, Stream Analytics, and Synapse, migrating on-premises data (Oracle, SQL Server, MongoDB) to Azure Data Lake Storage (ADLS) via ADF v1/v2 and PowerShell.
- Engineered scalable ETL workflows with Spark SQL, U-SQL, and T-SQL for data ingestion, transformation, and loading, automating pipelines via Azure Data Factory (Copy Activity, Custom Pipelines) and optimizing performance through parallel execution.
- Architected near-real-time data pipelines using Apache Kafka, Spark Streaming, and HDFS to process clickstream data for personalized user experiences and behavioral insights.
- Automated ETL processes by designing DAGs (Directed Acyclic Graphs) and Oozie workflows, while developing Python-based configurable pipelines for scheduled updates to customer-facing systems.
- Migrated legacy databases (DB2, SQL Server) to Azure Data Lake and Blob Storage, curating raw data into analytics-ready formats using Azure Databricks and Spark.
- Collaborated with stakeholders to analyze system enhancements, perform impact analysis for ETL changes, and translate business requirements into technical specifications.
- Utilized PySpark and Scala for high-volume ETL tasks, including data mapping and transformations, while processing streaming data from Kafka clusters into staging areas for downstream analytics.
- Exported processed data to relational databases via Sqoop and created interactive Tableau dashboards for sales performance tracking and executive reporting.
- Managed Spark clusters to aggregate daily sales data, optimize resource allocation, and generate leadership reports.
- Implemented ad-hoc analysis solutions using Azure Data Lake Analytics and HDInsight for rapid business insights and exploratory data science.
- Ingested raw data into Azure SQL DW, Blob Storage, and Data Lake while enforcing governance policies for structured/unstructured data consistency.
- Developed Spark jobs to process streaming data, load staging tables, and orchestrate workflows for BI team consumption.
- Worked on developing ETL Workflows on the data obtained using Scala for processing it in HDFS and HBase using Oozie

Environment: Kafka, Impala, PySpark, Azure, HDInsight, Data Factory, Databricks, Data Lake, Apache Beam, Cloud Shell, Tableau, Cloud SQL, MySQL, Posgres, SQL Server, Python, Scala, Spark, Hive, Spark-SQL, No SQL, MongoDB, TensorFlow, Jira.

Fifth Third Bank - Cincinnati, Ohio
Senior Data Engineer

Aug 2021 - Aug 2023

Responsibilities:

- Led requirement gathering, design, development, testing, and post-deployment support for data solutions. Collaborated with QA teams to execute SIT/UAT and ensure seamless production releases.
- Built and optimized transformation processes using Hadoop, Oracle, PySpark, and Hive. Focused on scalability and efficiency for high-volume enterprise data.
- Engineered ETL workflows with Azure Data Factory, Databricks, and Synapse Analytics. Migrated on-premises data (SQL Server, Oracle) to Azure ADLS and Blob Storage.
- Partnered with Solution Architects and DevOps teams to align technical deliverables with business goals. Translated stakeholder needs into actionable technical specifications.
- Managed code repositories using GitHub for collaborative development. Automated CI/CD pipelines to streamline deployments and reduce manual errors.
- Developed ETL jobs to extract Salesforce data and load it into Azure Synapse data marts. Ensured data consistency and accessibility for BI reporting.
- Optimized Spark SQL queries and DataFrames for performance-critical transformations. Leveraged RDDs and custom PySpark scripts for complex data processing.
- Processed streaming data using Kafka, Spark Streaming, and Azure Event Hubs. Stored results in Parquet format on ADLS for downstream analytics.
- Automated legacy database migrations (DB2, SQL Server) to Azure using Sqoop and ADF. Reduced downtime and ensured data integrity during transitions.
- Wrote shell scripts to automate repetitive tasks and scheduled cron jobs. Improved pipeline reliability and reduced operational overhead.
- Designed analytics-ready data marts in Azure Synapse for business teams. Created Tableau dashboards to visualize sales performance and KPIs.
- Practiced Scrum with daily standups, sprint planning, and retrospectives. Delivered iterative solutions aligned with evolving business needs.
- Scheduled and monitored database jobs using Azure Data Factory pipelines. Ensured timely execution of critical business processes.

- Enforced quality checks and validation rules during ETL processes. Reduced data inconsistencies by 25% across pipelines.
- Researched and tested new integration tools (e.g., Apache Beam) to enhance team capabilities. Accelerated data processing workflows by 20%.
- Worked with QA teams to define test cases and resolve defects. Ensured compliance with functional and non-functional requirements.
- Converted Kafka streams into structured DataFrames for real-time analytics. Enabled personalized user experiences through behavioral insights.
- Prepared technical documentation and user guides for deployed solutions. Conducted training sessions to empower end-users and support teams.
- Successfully loaded files to Azure Synapse, ADLS, and Hive from Oracle and SQL Server using Azure Data Factory and Sqoop.

Environments: Azure, Hadoop, Map Reduce, Hive, Sqoop, Spark, Parquet, Oracle12c, SQL Plus, SQL Loader, SQL Developer, PL/SQL, Kafka, GitHub, Shell Scripts, UNIX, PySpark, Databricks, Synapse Analytics, ADLS, Blob Storage, Azure Data Factory

Merck Pharmacy - Branchburg, NJ
Data Engineer

March 2019 - July 2021

Responsibilities:

- Designed batch and streaming pipelines using PySpark and AWS Glue to ingest structured/unstructured data from APIs, IoT devices, and S3. Enforced data quality through schema validation, deduplication, and Great Expectations framework integration.
- Built Kafka Connect pipelines to stream real-time transactional data into AWS Kinesis and transformed it using Spark Structured Streaming. Automated infrastructure with Terraform to deploy EC2 Auto Scaling Groups and EMR clusters with Spot Fleet cost optimization.
- Provisioned multi-region AWS architecture (VPCs, IAM roles, S3 buckets) using Terraform modules and CloudFormation stacks. Integrated Ansible for configuration management to standardize AMIs and reduce deployment drift.
- Automated Lambda-triggered workflows using S3 Event Notifications and CloudWatch Events. Scaled ECS Fargate tasks dynamically based on Application Load Balancer metrics.
- Enforced least-privilege IAM policies with SCP guardrails and AWS Organizations. Audited access using AWS Config and CloudTrail to ensure compliance with SOC2 and GDPR.
- Deployed VPC peering and Transit Gateway for hybrid cloud connectivity between on-prem data centers and AWS. Secured traffic with Network ACLs, Web Application Firewall (WAF), and AWS Shield for DDoS protection.
- Designed DynamoDB tables with Global Secondary Indexes (GSIs) for low-latency queries on user session data. Implemented Time-to-Live (TTL) policies to auto-archive stale data to S3 Glacier.
- Orchestrated Step Functions with Lambda and Glue Jobs for multi-stage ETL workflows. Monitored execution with X-Ray tracing and CloudWatch Service Lens for end-to-end visibility.
- Configured CloudWatch Alarms for EC2 CPU Utilization and Lambda Throttles, triggering SNS alerts and PagerDuty integrations. Built Grafana dashboards to visualize Prometheus metrics from EKS clusters.
- Tuned Spark jobs using Kyro Serialization and off-heap memory to reduce garbage collection pauses. Leveraged Delta Lake with Z-Ordering to optimize S3 file scan performance by 50%.
- Applied S3 Object Lock for write-once-read-many (WORM) compliance and AWS Macie for PII detection. Partitioned Parquet datasets by date/hour to accelerate Athena and Redshift Spectrum queries.
- Deployed Kafka brokers on EC2 with Apache ZooKeeper for cluster coordination. Streamed clickstream data into Spark Structured Streaming for real-time sectorization and anomaly detection.
- Transformed JSON/CSV in S3 using Glue Dynamic Frames and Python Shell jobs. Loaded curated data into Redshift with RA3 nodes and AQUA for ML-powered query acceleration.
- Optimized Redshift workloads using WLM query queues and Concurrency Scaling. Implemented Hive LLAP and Tez to reduce Hive query latency by 35%.
- Onboarded Salesforce data via AWS AppFlow into S3 and transformed it into Iceberg tables for ACID-compliant analytics. Integrated MongoDB Atlas with AWS Glue for CDC replication.
- Migrated Teradata fact tables to HDFS using Sqoop with --split-by parallelism. Ingested log files via Flume agents into Hive partitions for daily batch reporting.
- Designed Kimball-style Star Schema in Erwin for Redshift, with fact/dimension tables and surrogate keys. Validated referential integrity using dbt tests and Redshift Stored Procedures.
- Resolved Spark data skew using salting and broadcast joins. Optimized HDFS with Erasure Coding and YARN Fair Scheduler for multi-tenant resource fairness.

Environments: Python3.8, Shell Scripting, PySpark, Spark3.0 (SQL, DF/DS), AWS (S3, Lambda, DynamoDB, Glue, Kinesis, Redshift, IAM, CloudWatch, Step Functions, Macie, EC2, EMR, SNS), Terraform/Ansible, Kafka, HDFS/Hive, Erwin, dbt, Grafana/Prometheus, Sqoop/Flume, UNIX/PowerShell, Delta Lake/Iceberg.

Sonata Software - Hyderabad, India
Data Engineer

April 2016 - Nov 2018

Responsibilities:

- Deployed and maintained AWS production systems (EC2, S3, EMR) to ensure high availability, scalability, and security. Automated infrastructure provisioning using CloudFormation and Terraform for repeatable environments.
- Migrated on-premises SQL Server data to AWS Redshift, designing optimized schemas and partitioning strategies. Integrated AWS Glue for ETL workflows and S3 for cost-effective storage of raw and transformed data.
- Designed Star/Snowflake schemas and Hive tables (static/dynamic partitions, bucketing) aligned with business requirements. Collaborated with stakeholders to refine OLAP/OLTP models for reporting and analytics.
- Prototyped ML workflows using AWS Sage Maker for predictive analytics. Automated transient EMR clusters for big data processing, leveraging PySpark for data transformations and Hive integrations.
- Developed PySpark scripts to process Parquet data and load it into Hive tables via Spark SQL. Wrote Python and Shell scripts to automate server management, backups, and security patches.
- Built CloudFormation templates to provision VPCs, subnets, and S3/Glacier storage with IAM policies. Standardized CI/CD pipelines for Spark jobs using Code Pipeline and version control.
- Processed large-scale log data using Hadoop (HDFS), Kafka, and PySpark RDDs. Transformed raw datasets into analytics-ready formats for downstream reporting.
- Enforced data security by sanitizing PII/PHI with Python MapReduce jobs. Configured S3 Object Lock and Glacier policies for compliance and audit readiness.
- Migrated on-premises applications to AWS using EC2 and EMR. Configured VPC peering and Direct Connect for secure hybrid data access and processing.
- Designed S3 lifecycle policies and archival strategies to reduce storage costs. Optimized Redshift clusters with workload management (WLM) and partitioning.
- Leveraged PySpark Data Frames and broadcast joins for efficient data aggregation. Exported results to OLTP systems using Sqoop for seamless integration.
- Managed S3 buckets with versioning, encryption, and access policies. Utilized Glacier for long-term archival and disaster recovery workflows.

Environment: AWS, AWS EMR, SQL, ETL, Redshift, Sage Maker, DevOps, HDFS, Hive, Kafka, Sqoop, Shell Scripting, Spark, Map Reduce, Python, CloudFormation, Terraform, AWS Glue, IAM, Code Pipeline, VPC, Direct Connect, Glacier.